

Range Updates and Sums

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to maintain an array of n values and efficiently process the following types of queries:

1. Increase each value in range $[a, b]$ by x .
2. Set each value in range $[a, b]$ to x .
3. Calculate the sum of values in range $[a, b]$.

Input

The first input line has two integers n and q : the array size and the number of queries.

The next line has n values $t_1, t_2, \dots, t_n$: the initial contents of the array.

Finally, there are q lines describing the queries. The format of each line is one of the following: "1 a b x ", "2 a b x ", or "3 a b ".

Output

Print the answer to each sum query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq t_i, x \leq 10^6$
- $1 \leq a \leq b \leq n$

Example

Input	Output
6 5 2 3 1 1 5 3 3 3 5 1 2 4 2 3 3 5 2 2 4 5 3 3 5	7 11 15